

Bangladesh University of Engineering and Technology

CSE208: Data Structure and Algorithms Sessional II

Online 3

Section: C1 + C2

Time: 35 Minutes

The Interstellar Gates

Humanity has colonized N planets in a 3D sector. The coordinates of the i -th planet are (x_i, y_i, z_i) .

To create a trade network, we need to connect all planets. The cost to build a warp gate between planet i and planet j is the minimum of the coordinate differences:

$$\text{Cost}(i, j) = \min(|x_i - x_j|, |y_i - y_j|, |z_i - z_j|)$$

Calculate the minimum cost to connect all planets so that every planet is reachable from every other planet.

Note: Because N can be large (100,000), $O(N^2)$ solution is NOT acceptable.

Input Format

- First line: Integer N .
- Next N lines: Three integers x, y, z describing a planet's location.

Output Format

Single integer: The minimum cost to connect all planets.

Constraints

- $1 \leq N \leq 100,000$
- $0 \leq x, y, z \leq 10^9$

Samples

Sample 1

Input	Output
2 0 0 0 1 10 20	1

Explanation: $Cost = \min(|0-1|, |0-10|, |0-20|) = 1$.

Sample 2

Input	Output
3 0 0 0 10 5 5 20 10 10	10

Sample 3

Input	Output
3 1 5 10 2 8 15 7 12 11	2